

FARO AP, Portugal

WMO: 085540

Lat: 37.0167N

Lon: 7.9719W

Elev: 8

StdP: 101.23

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	6.7	-4.4	2.6	11.2	-1.7	3.3	11.8	12.6	14.6	11.2	14.6	2.3	20	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	32.0	19.9	30.4	19.9	29.1	19.7	22.5	27.5	22.0	26.9	21.4	26.2	4.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	15.8	24.2	20.5	15.2	23.8	19.9	14.6	23.6	66.2	27.5	64.1	26.8	62.2	26.2	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.3	8.2	3.0	36.1	1.4	2.3	2.0	37.7	1.1	39.1	0.3	40.4	-0.7	42.1		
(5)			0.9	23.8	1.8	0.7	-0.4	24.3	-1.4	24.7	-2.4	25.1	-3.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	12.4	12.9	14.9	16.5	19.2	22.3	24.3	24.5	22.2	19.8	15.7	13.5	(6)
(7)		DBStd	4.84	2.11	2.07	2.01	2.28	2.76	2.60	2.41	2.37	2.16	2.19	2.41	2.13	(7)
(8)		HDD10.0	9	4	3	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD18.3	777	183	152	109	64	22	1	0	0	1	10	84	151	(9)
(10)		CDD10.0	3006	80	84	151	195	284	369	443	448	367	304	171	109	(10)
(11)		CDD18.3	733	0	0	1	9	48	121	184	190	118	56	5	0	(11)
(12)		CDH23.3	4597	0	0	4	34	218	725	1414	1494	566	137	3	0	(12)
(13)		CDH26.7	1152	0	0	1	2	39	164	409	424	98	13	0	0	(13)
(14)	Wind	WSAvg	3.7	3.5	3.8	4.1	4.1	4.0	3.9	3.6	3.4	3.4	3.4	3.4	3.7	(14)
(15)	Precipitation	PrecAvg	502	79	56	52	35	22	11	0	2	20	62	80	76	(15)
(16)		PrecMax	881	310	194	175	125	69	71	10	23	118	185	229	283	(16)
(17)		PrecMin	267	0	2	0	0	0	0	0	0	0	0	2	8	(17)
(18)		PrecStd	177	74	48	40	36	18	15	2	5	30	52	60	76	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.5	20.2	23.3	26.6	30.2	32.7	34.9	34.2	30.7	28.2	23.2	20.0	(19)
(20)			MCWB	13.6	14.3	14.9	16.8	17.6	19.3	19.5	20.2	19.9	18.7	17.2	15.7	(20)
(21)		2%	DB	17.9	18.7	21.2	24.0	27.4	30.1	32.0	31.8	29.1	26.1	21.9	18.8	(21)
(22)			MCWB	14.3	14.1	14.5	15.2	17.4	19.0	20.0	20.3	20.0	18.7	17.1	15.9	(22)
(23)		5%	DB	17.1	17.6	20.0	22.2	25.7	28.4	30.2	30.3	27.8	24.8	20.9	18.0	(23)
(24)			MCWB	14.0	13.8	14.3	15.2	16.9	18.8	19.8	20.4	19.8	18.5	16.9	15.6	(24)
(25)		10%	DB	16.4	16.8	18.9	20.8	24.0	27.0	28.8	29.0	26.4	23.6	19.9	17.2	(25)
(26)			MCWB	13.5	13.5	14.1	15.0	16.7	18.7	19.7	20.2	19.5	18.5	16.5	14.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	16.4	17.4	18.5	20.6	22.1	22.9	23.5	22.7	21.7	20.3	17.9	(27)
(28)			MADB	17.7	17.8	19.9	23.0	25.5	27.3	28.3	28.3	26.8	24.0	21.7	18.5	(28)
(29)		2%	WB	16.0	15.6	16.4	17.5	19.4	21.2	22.2	22.7	22.0	20.9	19.0	17.2	(29)
(30)			MADB	17.0	17.2	18.6	21.1	23.7	26.2	27.8	27.8	25.9	23.1	20.6	18.1	(30)
(31)		5%	WB	15.1	15.0	15.8	16.8	18.6	20.6	21.6	22.2	21.4	20.4	18.1	16.4	(31)
(32)			MADB	16.3	16.7	18.0	19.9	22.8	25.6	27.2	27.3	25.2	22.5	19.8	17.5	(32)
(33)		10%	WB	14.2	14.3	15.2	16.1	18.0	19.9	21.0	21.6	20.8	19.8	17.3	15.5	(33)
(34)			MADB	15.9	16.2	17.6	19.2	22.1	24.9	26.6	26.7	24.6	22.1	19.1	16.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	7.5	7.5	7.6	7.7	8.1	8.8	8.3	7.3	6.8	7.3	6.9	(35)
(36)			MCDBR	7.6	7.9	9.2	9.5	10.0	10.2	10.9	10.1	8.8	8.0	7.4	6.7	(36)
(37)			MCWBR	5.3	5.3	5.5	5.3	5.1	5.2	5.5	5.3	4.9	4.8	4.7	4.8	(37)
(38)		5% WB	MCDBR	5.9	6.6	7.4	7.6	7.8	8.3	9.0	8.3	6.8	5.8	5.8	5.4	(38)
(39)			MCWBR	4.9	4.9	5.1	5.1	4.7	4.8	5.4	5.2	4.6	4.3	4.3	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.352	0.390	0.384	0.393	0.397	0.379	0.399	0.394	0.378	0.350	0.339	(40)	
(41)		taud	2.501	2.456	2.347	2.389	2.389	2.395	2.430	2.388	2.424	2.472	2.505	2.497	(41)	
(42)		Ebn at Noon	854	880	873	898	890	883	897	872	859	840	828	821	(42)	
(43)		Edh at Noon	81	95	116	118	120	119	114	116	107	93	81	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.60	3.56	4.77	5.97	7.05	7.73	7.92	7.04	5.56	3.95	2.84	2.31	(44)	
(45)		RadStd	0.28	0.39	0.42	0.38	0.44	0.37	0.20	0.21	0.30	0.30	0.26	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page